# CS 272: Final Project

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## Custom Environment:
## Car Accident

**GitHub Link:** https://github.com/niyatiagg/cs-272-final-project

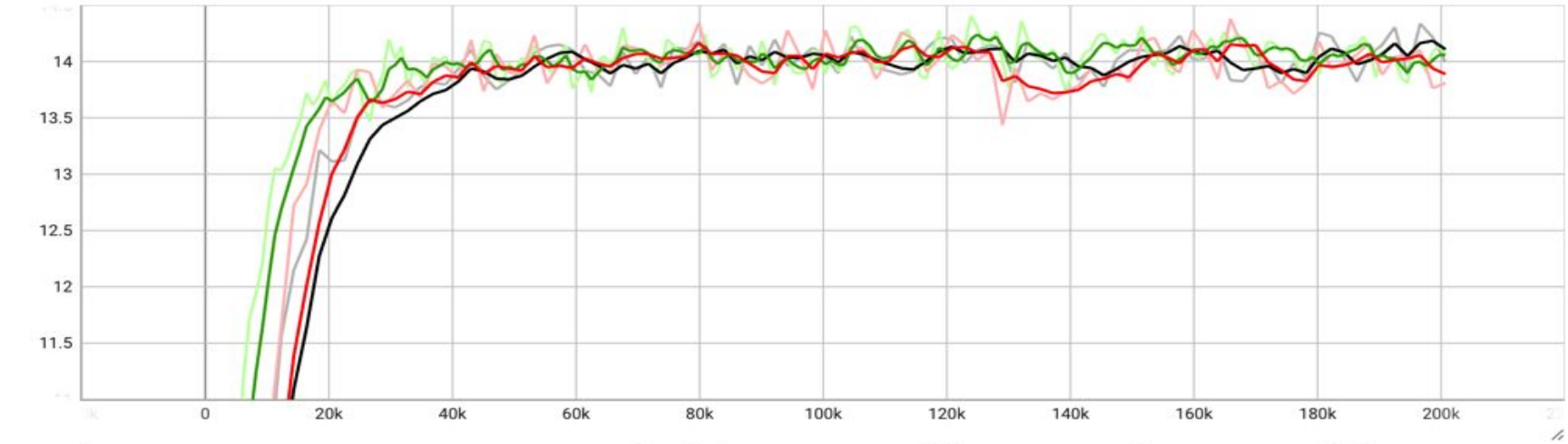
# Task 1- Merge Env

LIDAR OBSERVATIONS

# DRL Design Choices

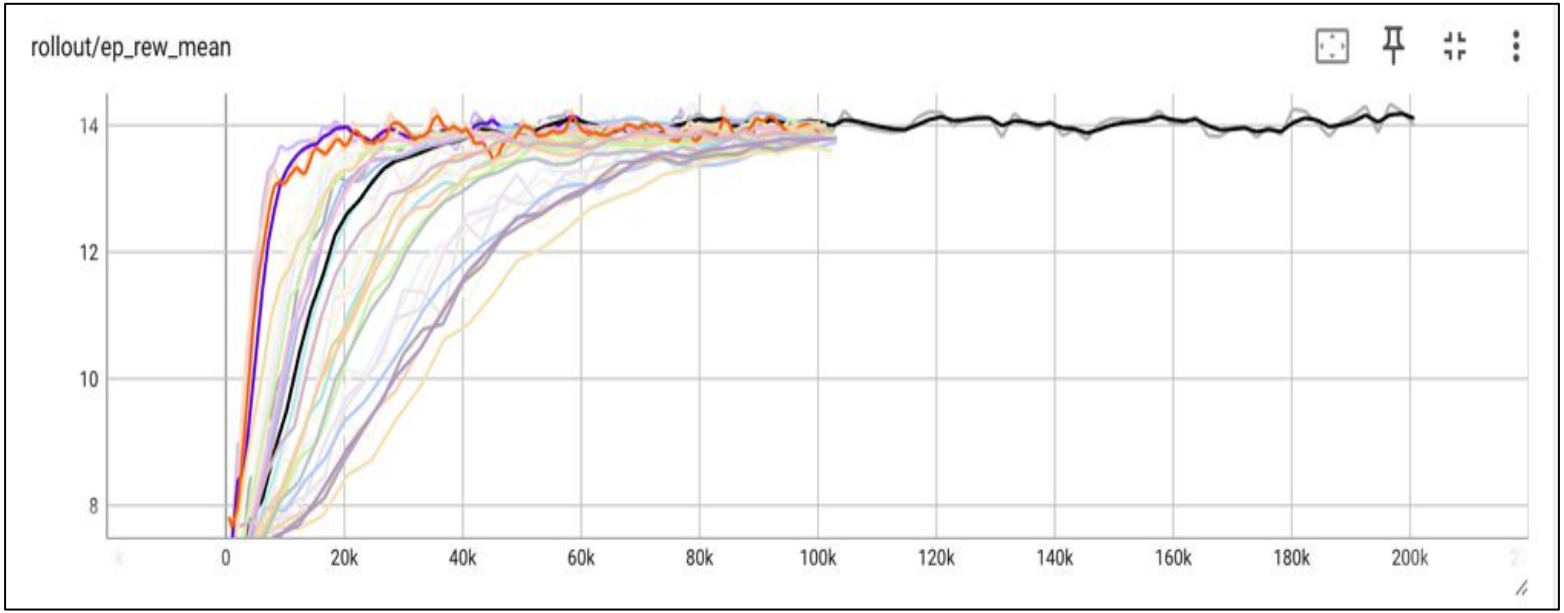
- **Baseline PPO from SB3** for **100k, 200k and 500k steps**
- **Manual Hyperparameter tuning**
  - Increasing the learning rate **[5e-4]**
  - Created a bigger DRL network architecture **[256, 256]**
- **Systematic hyperparameter tuning using Optuna**
  - 20 trials of 100k steps each
  - Improved RL Algorithm: PPO + Optuna best hyperparameters
- **Different exploration methods**:
  - Variation 1: PPO + High Entropy **[0.02]**
  - Variation 2: PPO + Entropy Decay **[0.02 -> 0]**
  - Variation 3: PPO + Parameter Noise **[Addition of Gaussian noise every 1k steps to policy network weights]**
- **Significantly Improved RL Algorithm : Reward Shaping**
  - Variation 1: PPO + Reward shaping **[Speed Bonus + Lane Penalty]**
  - Variation 2: PPO + Reward Shaping + Parameter Noise **[Addition of Gaussian noise every 1k steps to policy network weights]**
  - Variation 3: PPO + Reward Shaping + Entropy Decay **[0.02 -> 0]**

# Manual Hyperparameter Tuning



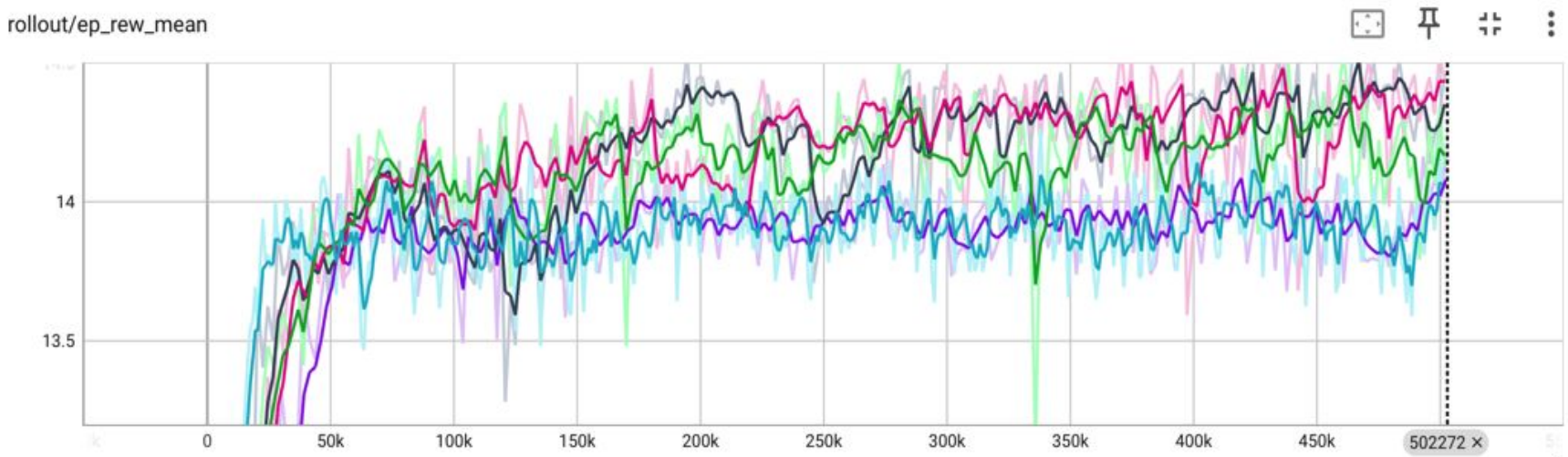

| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| manual_baseline_200k_1 | 14.108 | 13.9943 | 200,704 | 1.052 hr |
| manual_fast_lr_200k_1 | 14.0598 | 14.0608 | 200,704 | 1.201 hr |
| manual_large_net_200k_1 | 13.8872 | 13.8103 | 200,704 | 1.214 hr |

# Systematic Hyperparameter Tuning - Optuna



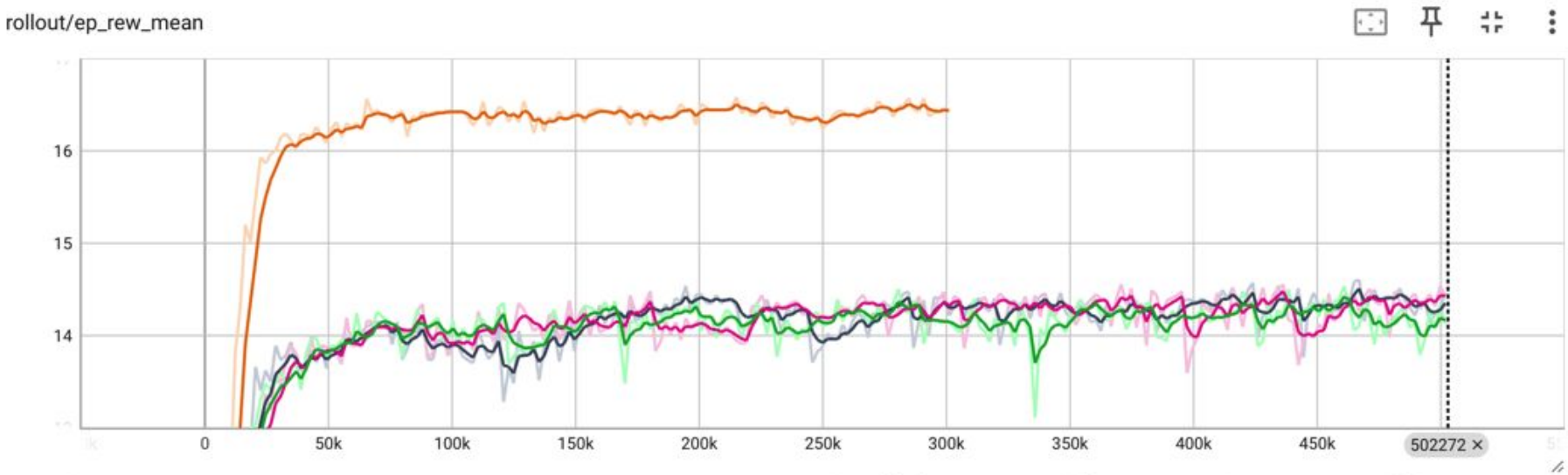

**Best was Trial 10**: lr = 4.06e-5; nsteps = 2304; batch size = 128; gamma = 0.964; gae = 0.955; clip_range = 0.142; entr_coeff = 0.0 (default)

# Optuna Best Param & Fast-Learning Models perform poorly




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| manual_baseline_mac_500k_seed42_1 | 14.3502 | 14.4602 | 501,760 | 3.746 hr |
| manual_fast_learning_mac_500k_seed42_1 | 14.0613 | 14.0491 | 500,736 | 3.145 hr |
| manual_large_network_mac_500k_seed42_1 | 14.4321 | 14.4289 | 501,760 | 4.55 hr |
| optuna_best_mac_500k_seed42_1 | 14.0796 | 14.1677 | 502,272 | 2.616 hr |
| ppo_high_entropy_mac_500k_seed42_1 | 14.1664 | 14.1281 | 501,760 | 2.939 hr |

# Reward Shaping Emerges as the Winner!




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| manual_baseline_mac_500k_seed42_1 | 14.3502 | 14.4602 | 501,760 | 3.746 hr |
| manual_large_network_mac_500k_seed42_1 | 14.4321 | 14.4289 | 501,760 | 4.55 hr |
| ppo_baseline_reward_shaping_300k_seed42_1 | 16.4365 | 16.4299 | 301,056 | 1.933 hr |
| ppo_high_entropy_mac_500k_seed42_1 | 14.1664 | 14.1281 | 501,760 | 2.939 hr |

# PPO + Reward-Shaping

**Speed Bonus**

- Extra positive reward proportional to the ego vehicle's current speed, capped at a target speed (e.g., 25 m/s).
- Purpose: Encourage the agent to drive faster instead of overly cautious or slow driving.

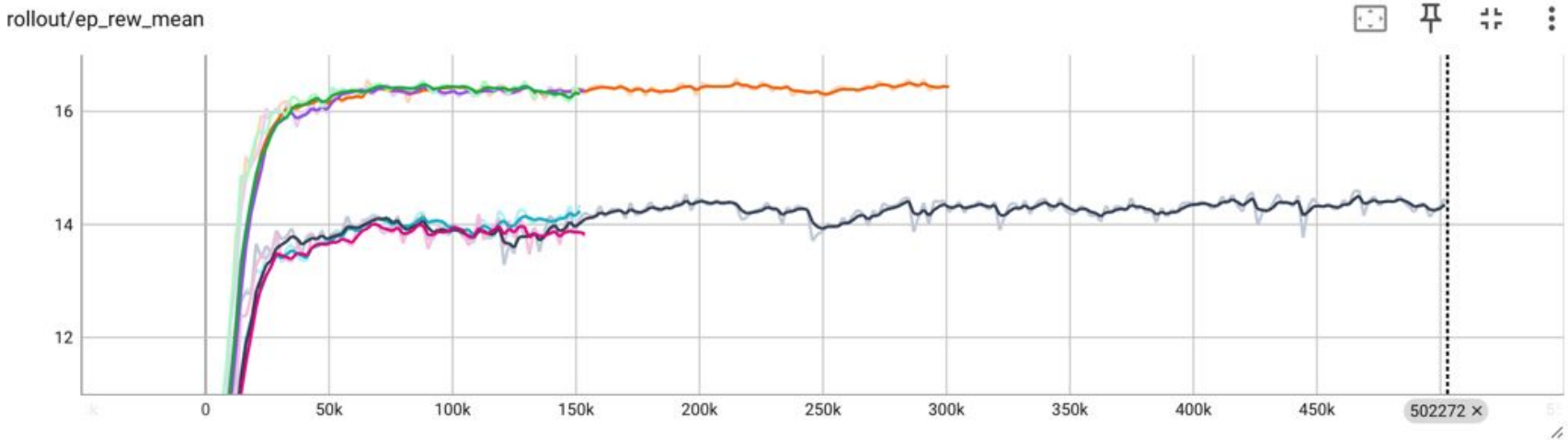
**Slow Speed Penalty**

- Penalize speed below a threshold (e.g., 10 m/s) to discourage stalling or waiting too long on merge ramps.
- Adds urgency to merging maneuvers

**Lane Change Penalty**

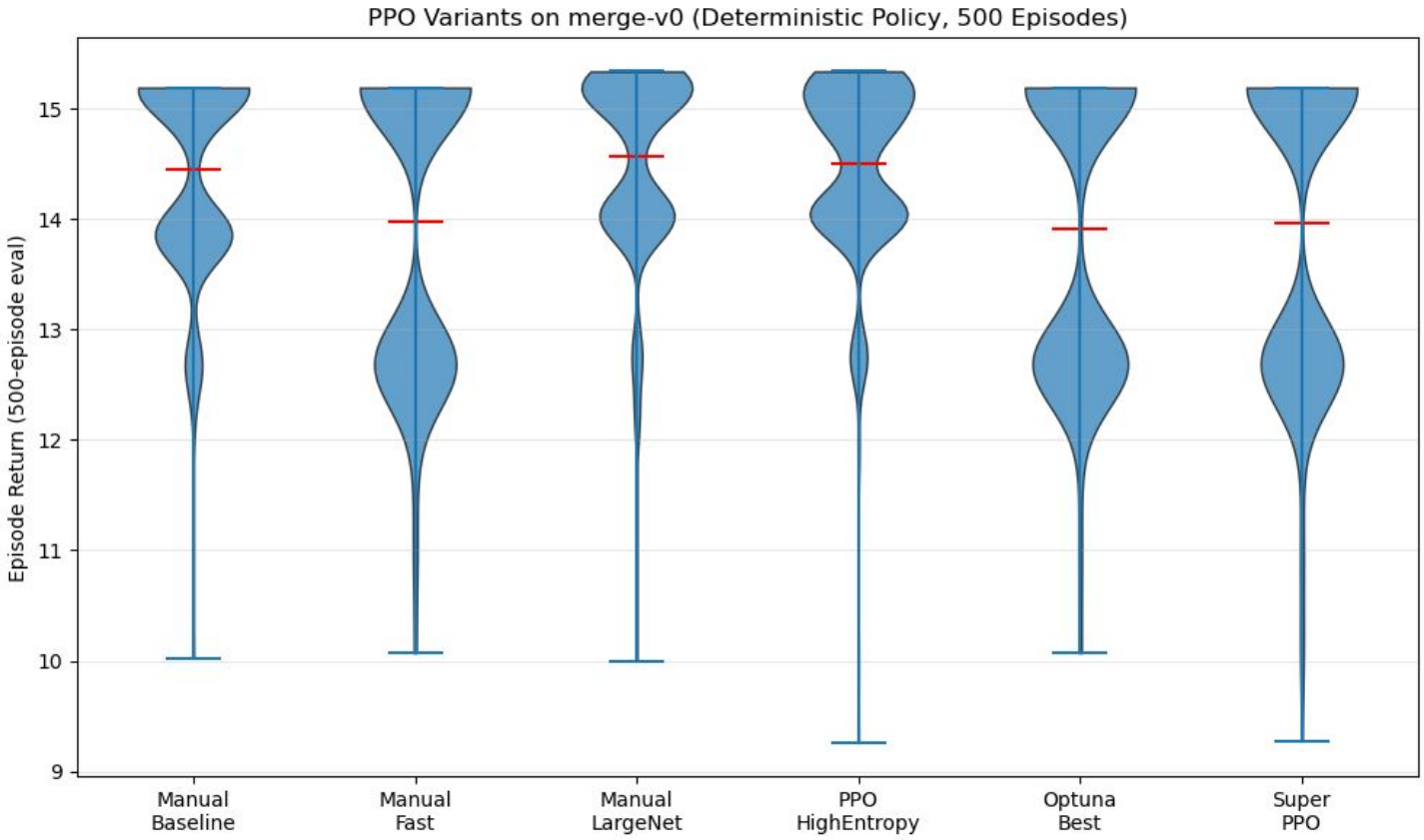
- Small negative reward when the agent changes lanes compared to the previous step.
- Purpose: Encourage smoother, fewer lane changes and avoid oscillations or unnecessary swerves.

# Variations of Reward-Shaping




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| manual_baseline_mac_500k_seed42_1 | 14.3502 | 14.4602 | 501,760 | 3.746 hr |
| ppo_baseline_entropy_decay_150k_seed42_2 | 14.2328 | 14.3504 | 151,552 | 46.93 min |
| ppo_baseline_param_noise_150k_seed42_0 | 13.8309 | 13.7809 | 153,600 | 48.43 min |
| ppo_baseline_reward_shaping_300k_seed42_1 | 16.4365 | 16.4299 | 301,056 | 1.933 hr |
| reward_shaping_entropy_decay_150k_seed42_1 | 16.3143 | 16.3026 | 151,552 | 1.621 hr |
| reward_shaping_param_noise_150k_seed42_0 | 16.367 | 16.3918 | 153,600 | 56.17 min |

# Task 1: Results on Merge (LidarObs) - Performance Test



# Our Custom Env

CAR ACCIDENT

# Task 2a: Custom Environment - Car Accident

**Car Accident Scenario**

- A 2-car crash on the highway, placed halfway down the road and spread across 2 lanes (adjacent to each other).
- Location of the crash is fixed, but the crash can occur in any 2 random lanes (adjacent to each other).
- A modified reward function with new rewards & penalties, suitable for a highway crash scenario.

**Objective of Ego-vehicle**

- Not colliding with other vehicles or the crashed vehicles.
- Moving away from the crash lanes.
- Adjusting speed as it approaches crash, but resuming driving at high speed, once away from the crash.
- Driving on the right-most lane.
- Not coming to a complete stop at any point.
- Not tailgating any vehicles at any point.
- Not driving off the road at any point.

# Task 2a: Custom Environment Setup

## Observation Space:

- Fixed-size vector, updated every step.
- Ego-centric: all positions/velocities are relative to the ego
- Normalized: values are scaled for stable learning.
- Observations : Lidar

## Action Space:

Discrete Meta Action

```
ACTIONS_ALL = {
        0: 'LANE_LEFT',
        1: 'IDLE',
        2: 'LANE_RIGHT',
        3: 'FASTER',
        4: 'SLOWER'
    }
```

## Reward Function

| Reward Type | Description |
|---|---|
| collision_reward | penalty for colliding with another vehicle |
| high_speed_reward | reward for driving at posted high speed limit |
| right_lane_reward | reward for driving on right-most lane |
| on_road_reward | reward for remaning on road |
| reaction_reward | penalty for being in crash lane(s) or close to crash |
| tailgating_reward | penalty for tailgating |
| job_well_done_reward | reward for safely avoiding crash & driving on right-most lane |

The **episode terminates** if -

1. The ego vehicle drives off the road.
2. The ego vehicle collides with another vehicle.

The **episode truncates** if -

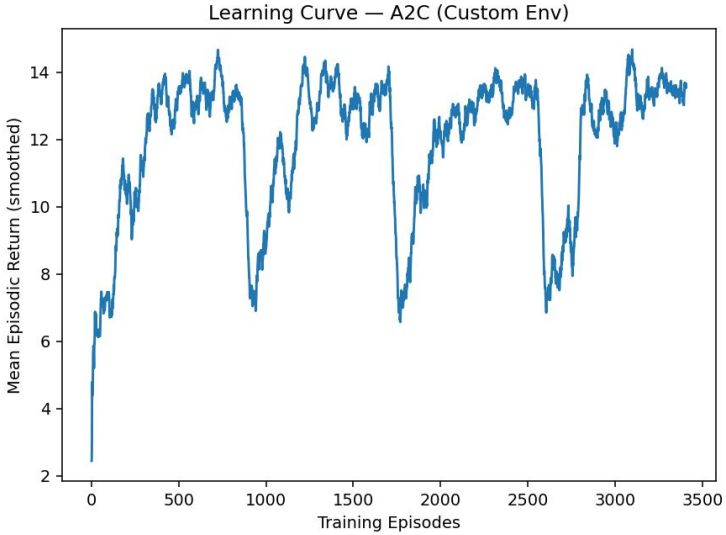
1. The time limit is exceeded.

# Task 2a: DRL Design Choices

- **Baseline** : Built agents from Stable Baseline 3 for 50k steps
  - First, built A2C - not very good performance
  - Then, compared two different SB3 models - PPO and TRPO
- **Optimised** : hyperparameter tuning using Optuna
  - Improved performance with agent trained by combining PPO + Optuna best hyperparameters.
  - 15 trials for (n_steps x n_envs x updates) steps and final training for best hyperparameters with ~ 400k steps
  - Added pruning for early stopping of weak trials to speed up training
  - To reduce the sawtooth dips and raise the plateau a bit more, increased samples per update.

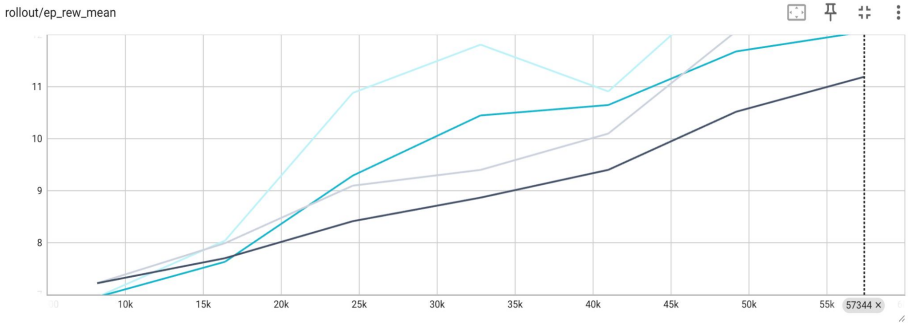
# Task 2a: Results on Custom Environment -A2C

- Shows A2C learning curve + violin: improves but dips often; outcomes spread wide; Baseline algorithm only—reveals need for steadier updates

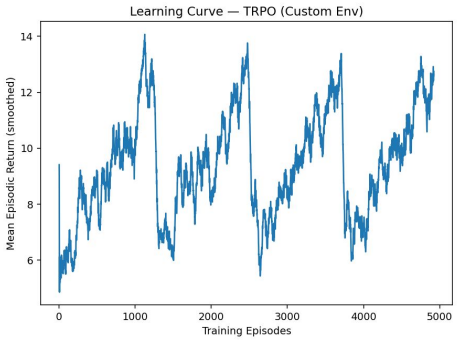


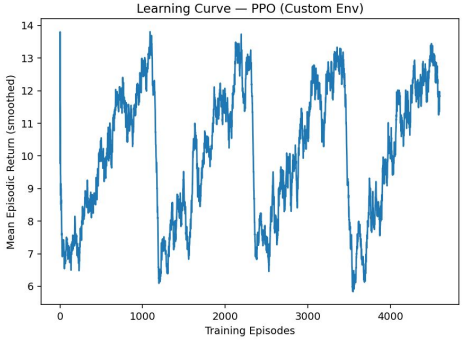
# Task 2a: PPO and TRPO

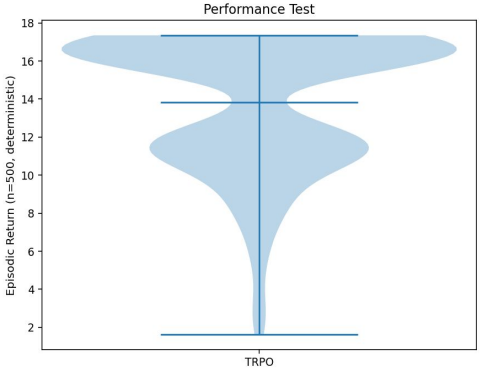
- both steadier than A2C; PPO typically rises faster; stability and average return improve over A2C

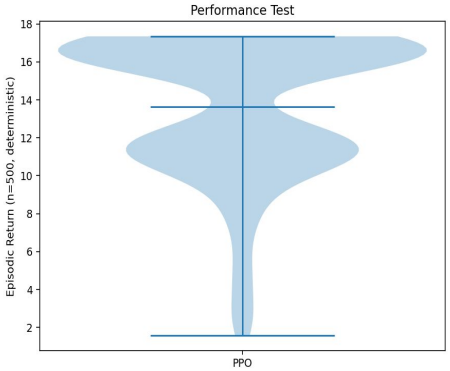



| Run | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| PPO_baseline_1 | 12.0561 | 12.5965 | 57,344 | 54.01 min |
| TRPO_baseline_1 | 11.1888 | 12.1467 | 57,344 | 53.24 min |



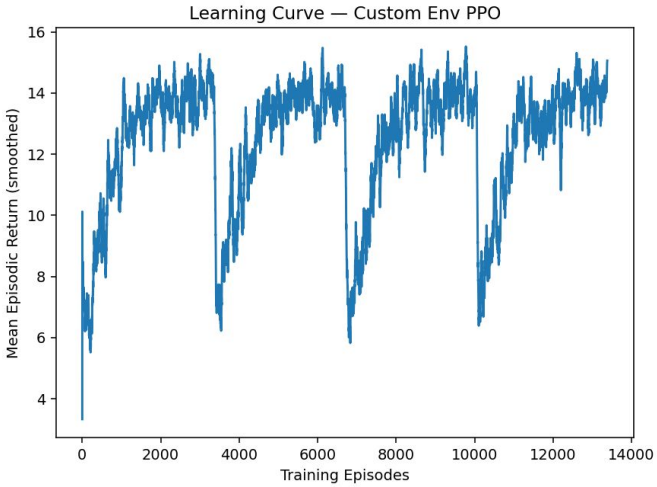



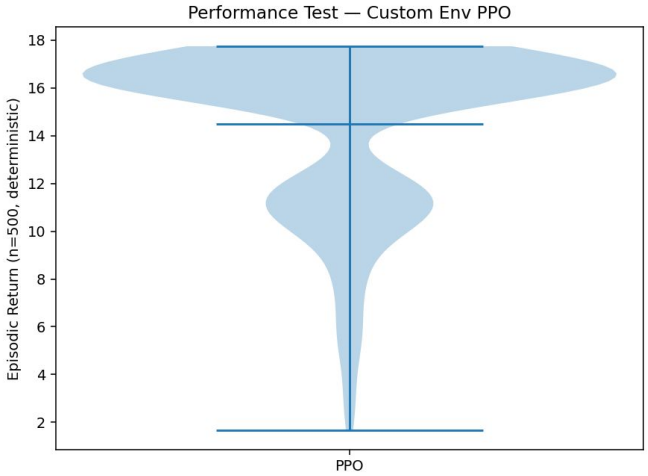



# Task 2a: Baseline PPO with timesteps=204_800

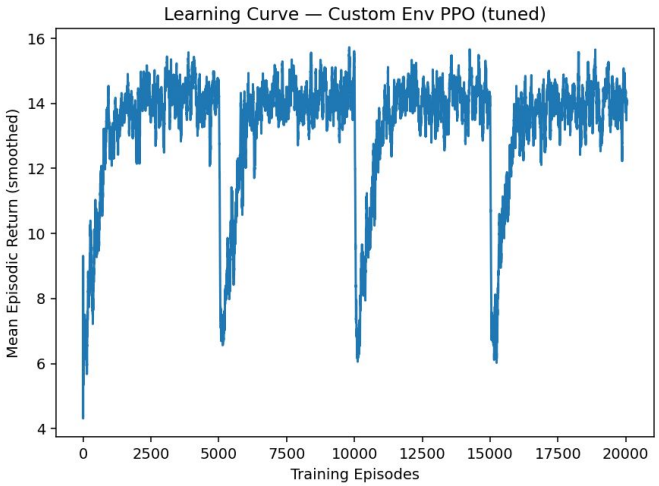
- Learning curve plateaus around 14–15 with periodic post-update dips; violin moderately tight.
- fixed PPO budget to establish a reference to beat in later slides

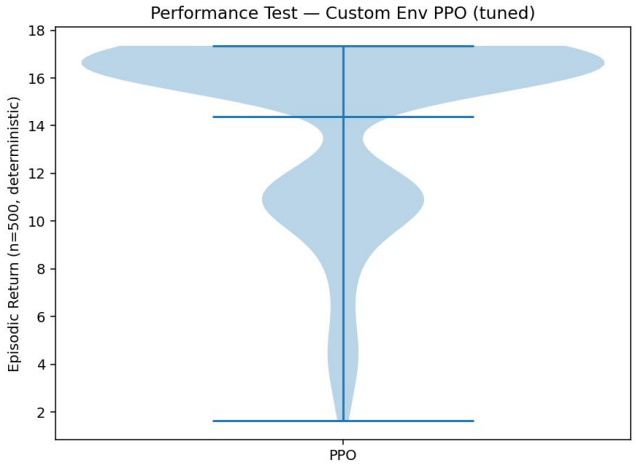


# Task 2a: Optimised PPO with optuna

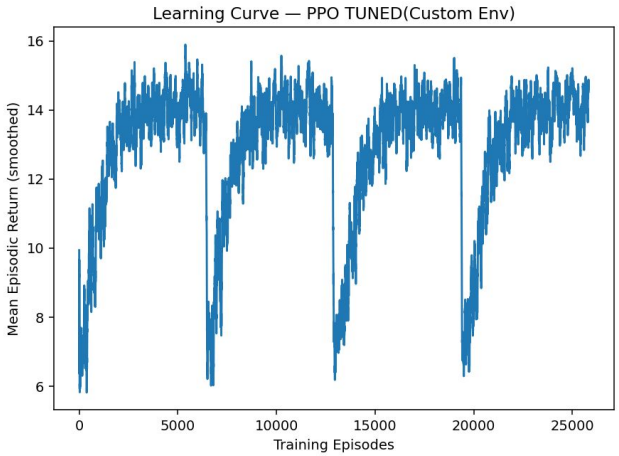
- Curve reaches high return sooner; dips shallower; violin tighter in terms of higher median; hyperparameter search (lr, batch, clip, γ, λ, etc.) lead to less noisy updates.

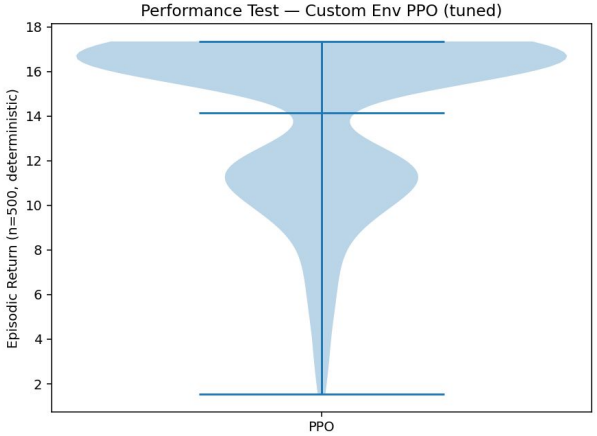


# Task 2a : Optimised PPO model + Manual Hyperparameter tuning

- Smoothest curve so far; violin comparable with last slide.
- bigger rollouts with mild schedules (lr/clip) on top of Optuna for extra stability



# Future Research

- *More training budget*: more Optuna trials and longer training to polish the best region.
- *Hybrid model idea*: I2A + PPO: another cool research idea but requires lots of time building a world-model + imagination features + update loop. Can be explored for better anticipation of hazards.
- *Harder scenarios*: partial visibility or changing traffic density to test robustness.
- *Richer reporting*: For our case, along with return, we can track success rate (cleared crash), collision rate, minimum time-to-collision, and lane violations so performance is easy to interpret.

# Video demo

```
OLE    TERMINAL    PORTS

python3 run_trained_custom_simple.py --live --episodes 3 --deterministic
```

# Appendix

-Highway (Lidar & Grayscale)
-Merge (Lidar* & Grayscale)
-Intersection (Lidar & Grayscale)
-Peer Group Custom Env

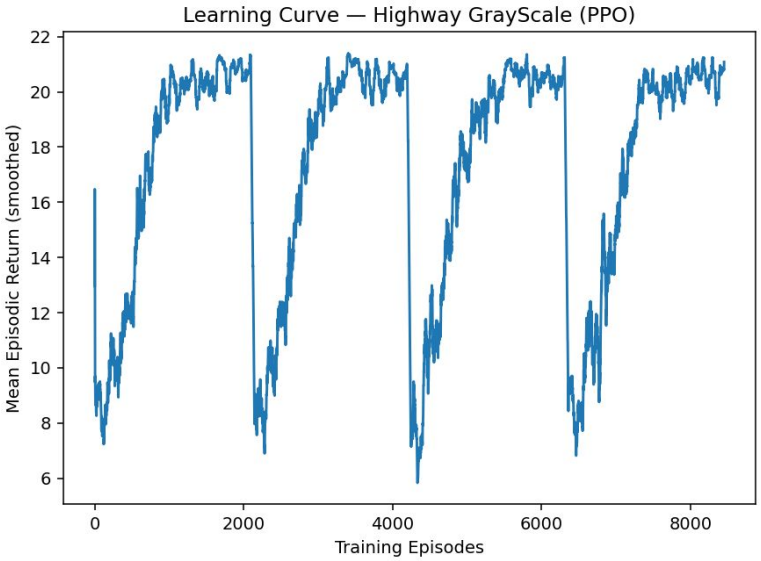
*Lidar Obs for Merge Env is shared in the beginning of the slide deck

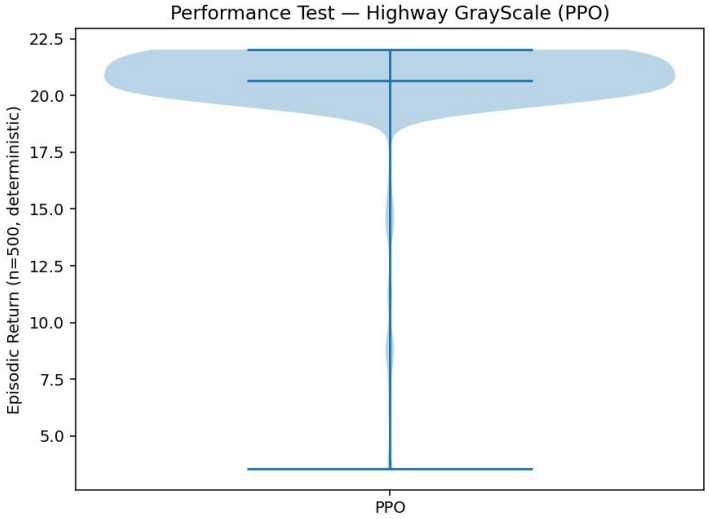
# Highway Env

Grayscale & Lidar

# Highway (Grayscale) - Learning curves and Performance Test

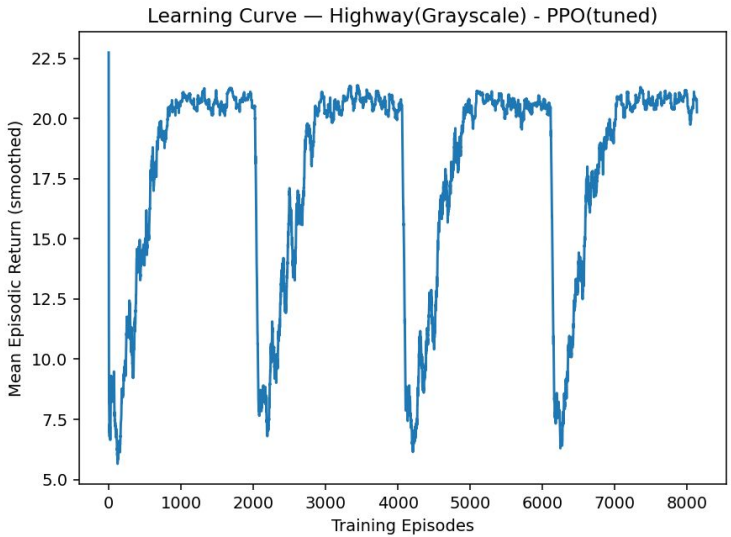
- Highway grayscale PPO learns quickly; returns cluster near the top band.

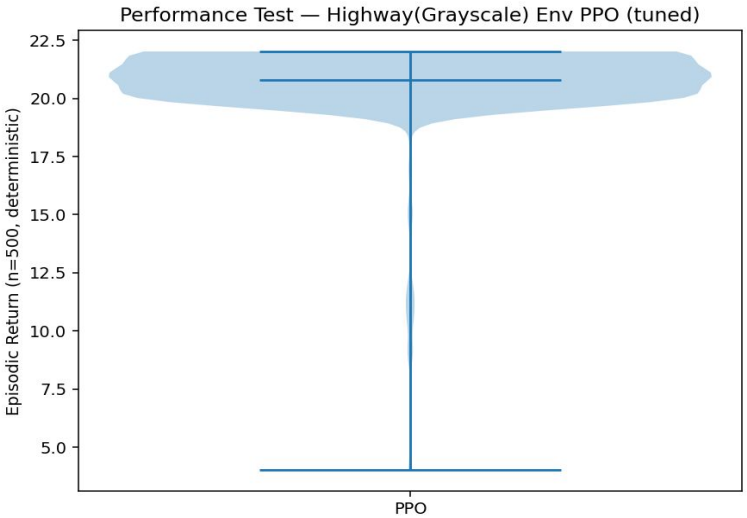


# Highway (Grayscale) - Optimized by Sys Hyperparam Tuning

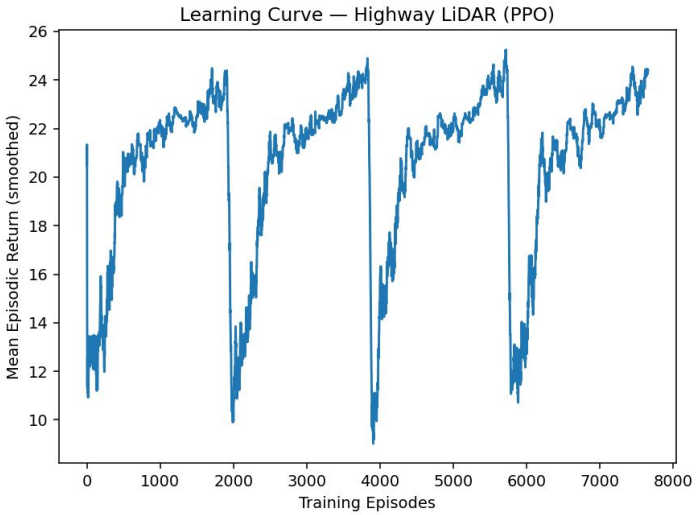
- Tuned curve is smoother with more time at peak; violin more concentrated (around 20–22).
- Optuna on grayscale highway setup gives higher, steadier performance

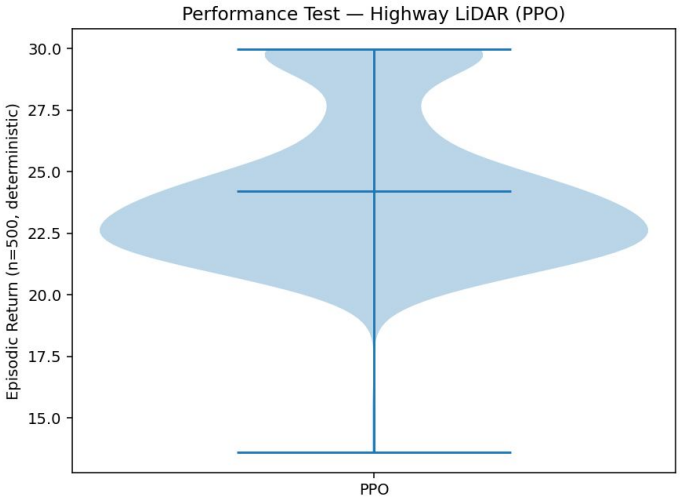


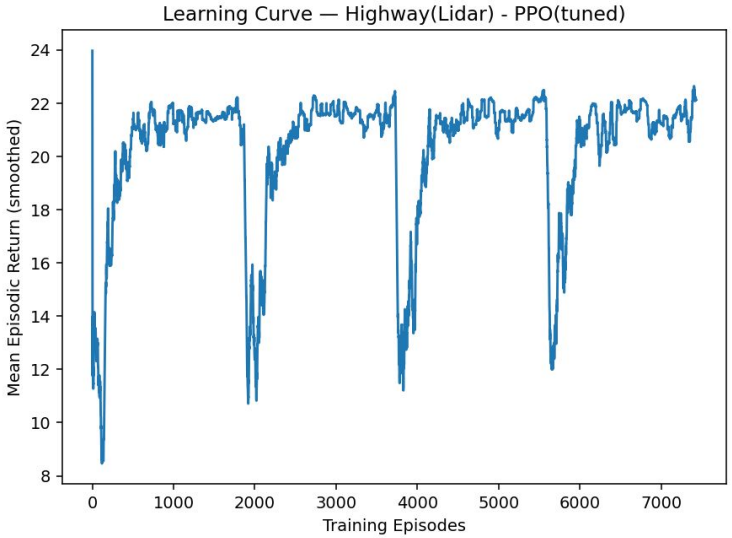
# Highway (LiDAR) - Learning curves and Performance Test

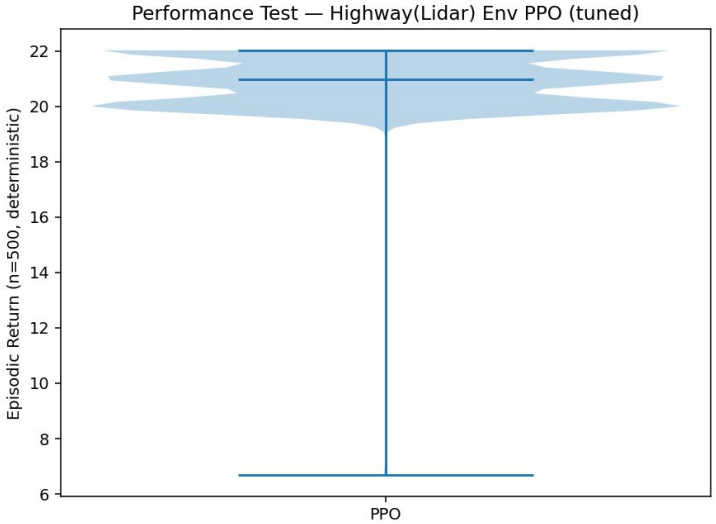
- With LiDAR vectors, PPO learns well but shows a bit more variance than grayscale.
- With obs type (LiDAR vs. images) alters learning dynamics/variance



- Tuning tightens the violin and smooths peaks; better median return than baseline LiDAR.
- Optuna for LiDAR config reduces noise and improves sample efficiency.



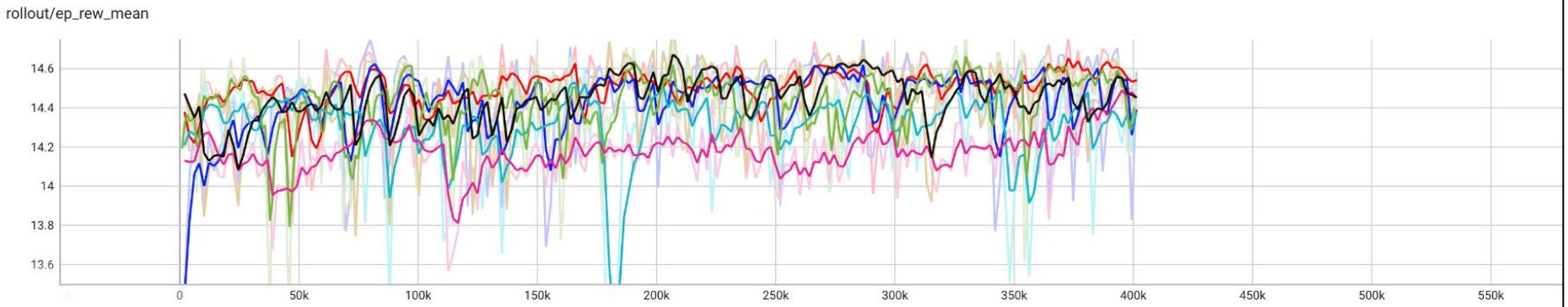
# Merge Env

-Grayscale Obs

# Merge Env (Grayscale Obs)- DRL Design Choices

- PPO from Stable Baseline 3 was chosen
- Improved Exploration
  - Used a fixed higher entropy coeff = 0.02
- Systematic Hyperparameter tuning (using Optuna)
  - 15 trials of 100k steps each
  - Trial 11 was the best!
- Changed Observation Shape
  - 84x84
  - 96x48
- Changed Stack Size
  - stack_size = 2
  - stack_size = 4
- Reward Shaping (using a wrapper script)

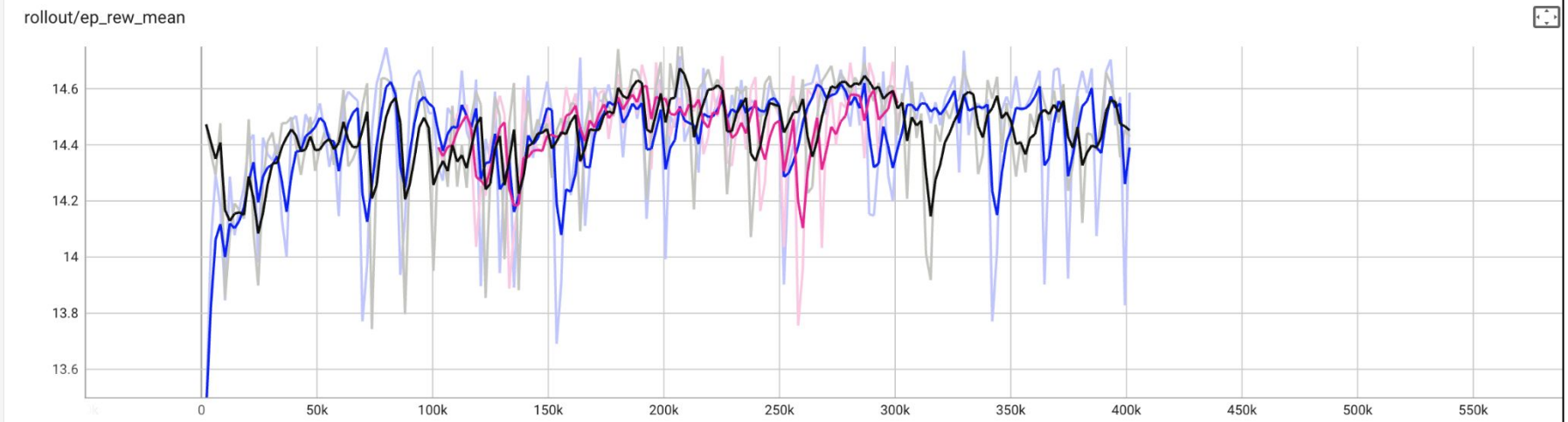
# Merge Env (Grayscale Obs) -Learning Curves




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| baseline_mps_500k_rerun_1 | 14.4514 | 14.4289 | 401,408 | 1.677 hr |
| merge_gray_96x48_500k_1 | 14.4506 | 14.4061 | 401,408 | 1.469 hr |
| merge_gray_ent_fixed_500k_1 | 14.3813 | 14.5233 | 401,408 | 1.548 hr |
| merge_gray_reward_improved_500k_1 | 14.3904 | 14.5864 | 401,408 | 1.55 hr |
| merge_gray_stack2_500k_1 | 14.5407 | 14.5517 | 401,408 | 1.51 hr |
| optuna_500k_3 | 14.4034 | 14.5019 | 400,384 | 1.799 hr |

Note: These models were initially run for 100k steps and then later run for 400k more steps. The timesteps of 0->400k actually refer to 100k->500k

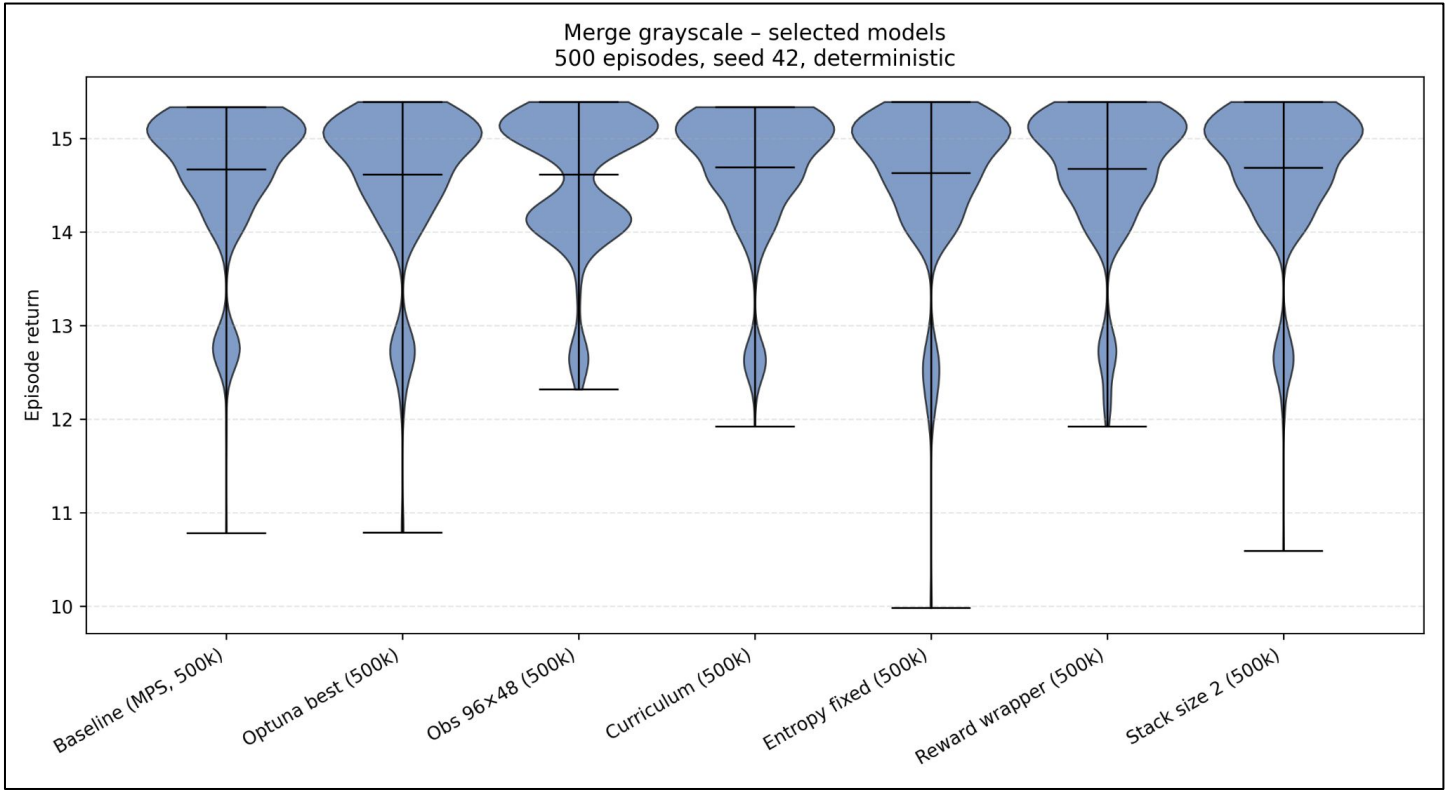
# Best Models - Reward Wrapper & Curriculum Learning




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| baseline_mps_500k_rerun_1 | 14.4514 | 14.4289 | 401,408 | 1.677 hr |
| merge_gray_reward_improved_500k_1 | 14.3904 | 14.5864 | 401,408 | 1.55 hr |
| merge_grayscale_curriculum/phase_2_0 | 14.5483 | 14.4776 | 301,056 | 29.32 min |

Note: These models were initially run for 100k steps and then later run for 400k more steps. The timesteps of 0->400k actually refer to 100k->500k

# Merge Env (Grayscale Obs) - Performance Test




**Reward wrapper and Curriculum learning were marginally better than the PPO Baseline.**

# Merge Env (Grayscale Obs) - Performance Test

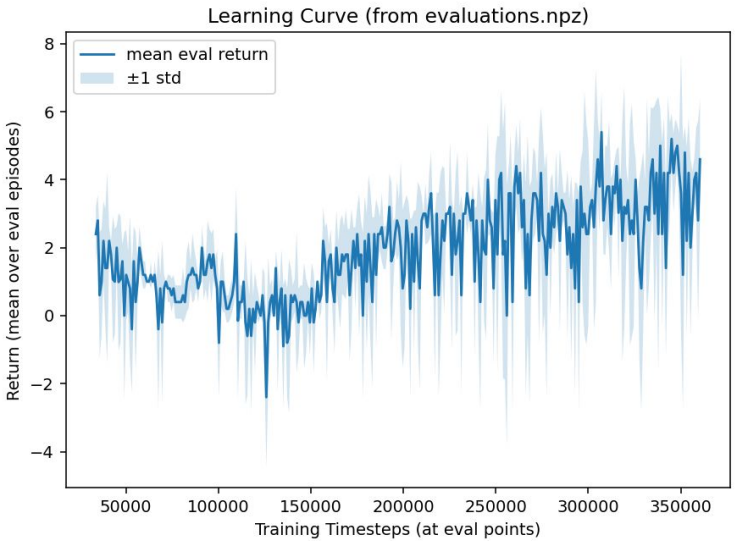
| Model | Mean | Median | StdDev | 25th% | 75th% | min | max | Time Steps |
|---|---|---|---|---|---|---|---|---|
| **Baseline PPO** | **14.67** | **15.01** | **0.73** | **14.53** | **15.11** | **10.79** | **15.34** | **500k** |
| **Optuna_Best** | 14.61 | 14.94 | 0.78 | 14.5 | 15.03 | 10.79 | 15.39 | 500k |
| **Obs Shape 96x48** | 14.61 | 14.98 | 0.72 | 14.17 | 15.17 | 12.32 | 15.39 | 500k |
| **Curriculum_Learning** | **14.69** | **15.01** | **0.70** | **14.58** | **15.11** | **11.93** | **15.34** | **600k** |
| **High Entropy = 0.02** | 14.63 | 15.01 | 0.78 | 14.53 | 15.11 | 9.98 | 15.39 | 500k |
| **Reward Wrapper** | **14.68** | **14.94** | **0.71** | **14.5** | **15.22** | **11.93** | **15.39** | **500k** |
| **Stack_size = 2** | 14.69 | 14.98 | 0.72 | 14.53 | 15.11 | 10.6 | 15.39 | 500k |

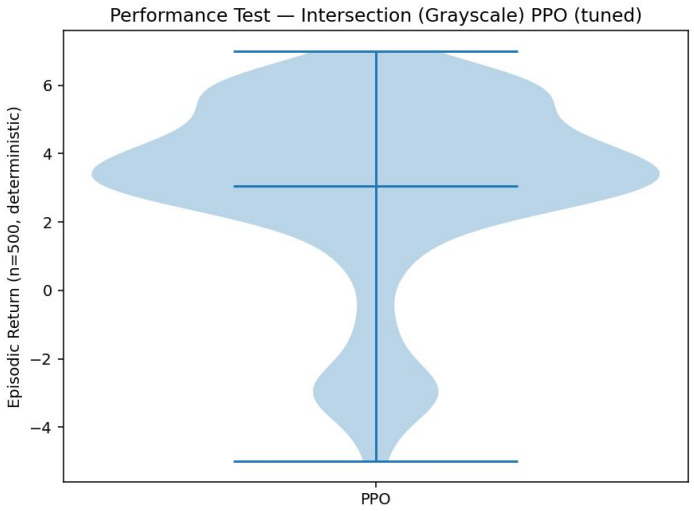
# Intersection Env

Lidar & Grayscale

# Intersection(Grayscale) - Learning Curve & Performance Test

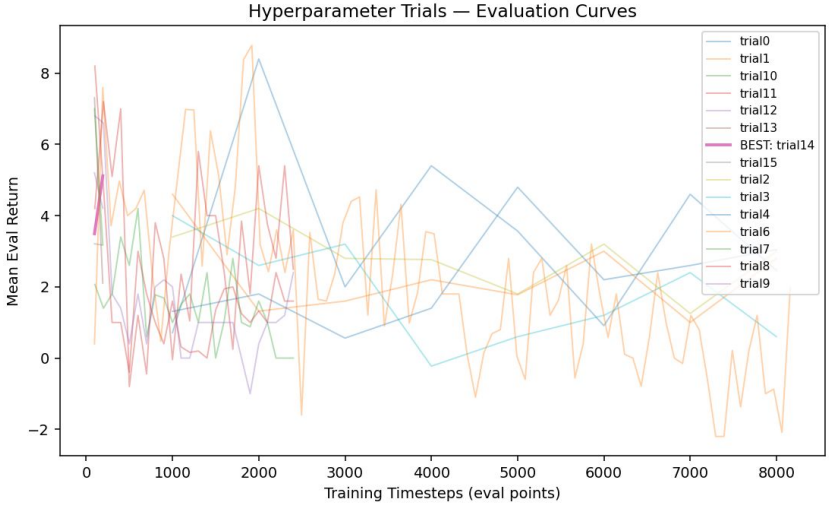
- Baseline PPO performance trained on more than 350k steps

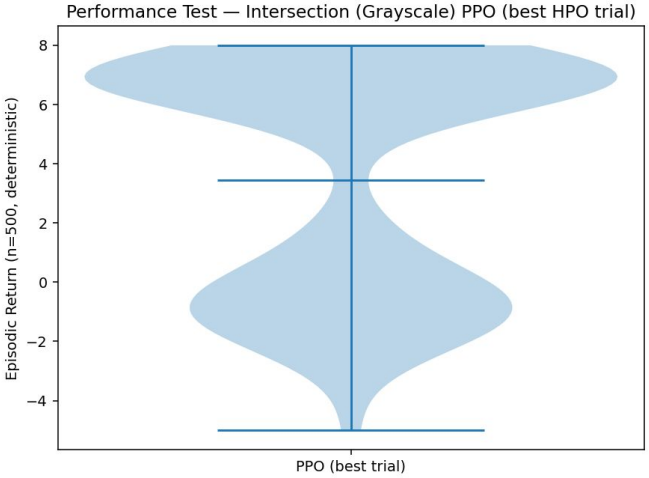


# Intersection (Grayscale) - Optimized Performance

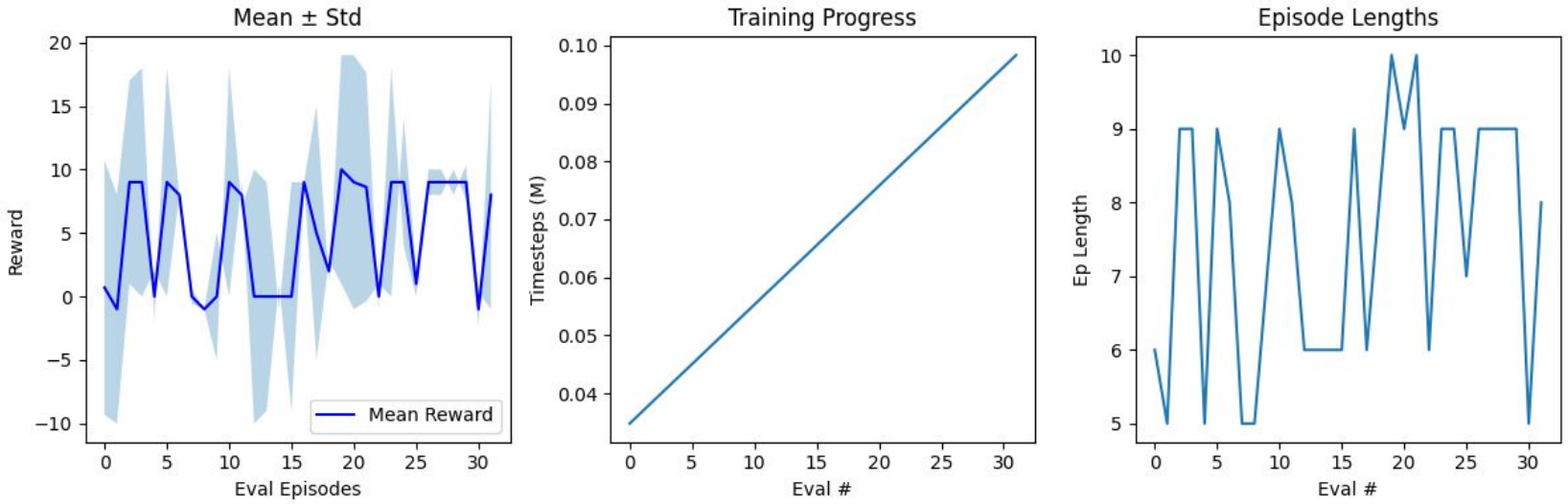
- Multiple HPO curves overlaid; best trial highlighted; violin shows its return spread.
- selection via HPO metrics (final/last-K/AUC), then visualize the winner's distribution.



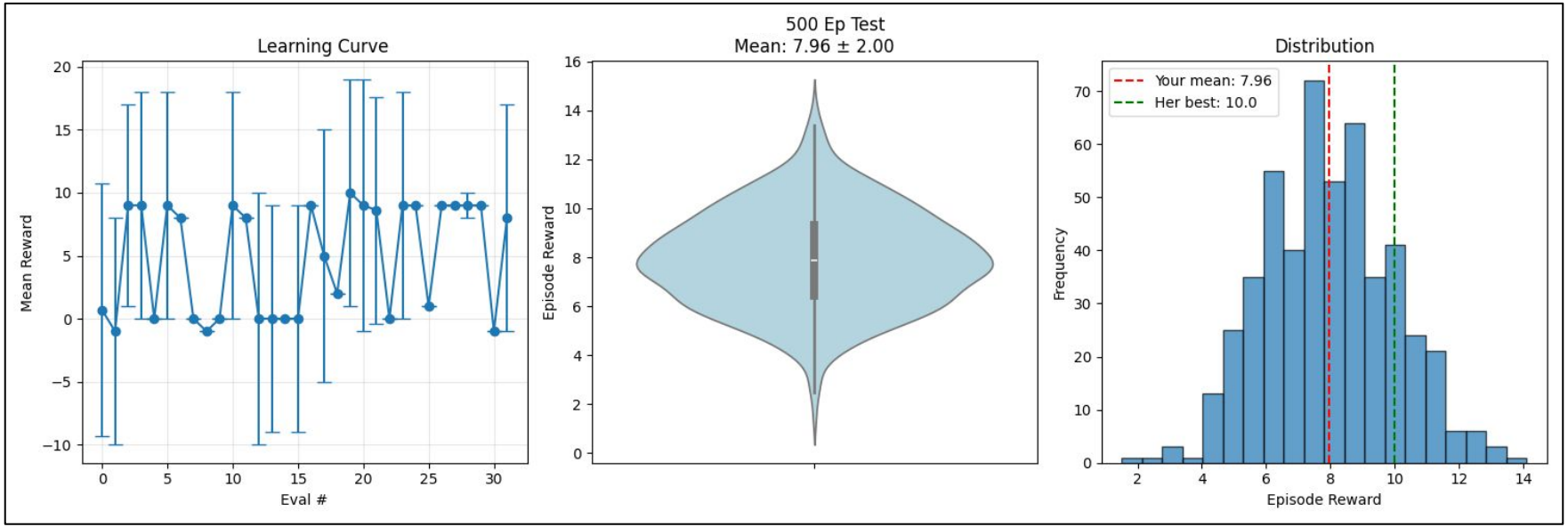



# Intersection Env (Lidar Obs) - Learning Curves

- Lillian ran a Baseline PPO model - trained for 32k steps.
- She ran a longer training that improved her Baseline PPO model
- Best mean reward: 10.00 ± 10.00; Final mean reward: 8.00 ± 9.00; Total timesteps: 98,304



# Intersection Env (Lidar Obs) - Performance Test




Best eval: 10.0
500-ep mean: 7.96 ± 2.00
Total timesteps: 98,304

# Peer Group
# Custom Env

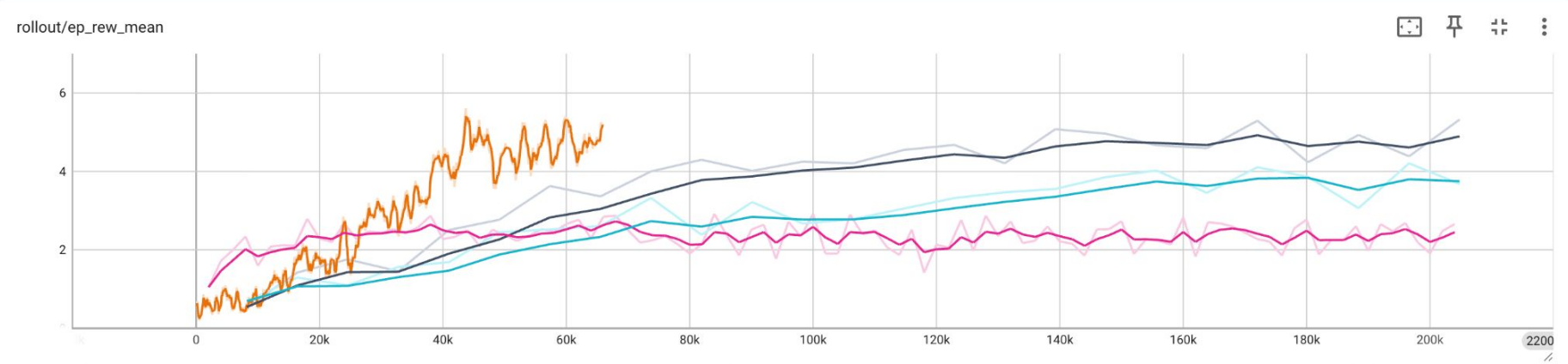
- We are Group 6
- Group 5 was our Peer Group

# Task 2b: Peer Custom Environment (Group 5)

- For the custom environment - we established baselines using A2C, TRPO PPO and DQN.
- From plots, we can see that PPO performs better than A2C and almost similar to TRPO but the tensorboard graph shows PPO as the clear winner
- Although DQN was actually showing most promise, it ran extremely slowly (fps=1), so we had to abort the training eventually.
- We decided to stick to PPO as our baseline model.

# Task 2b : DRL Design Choices

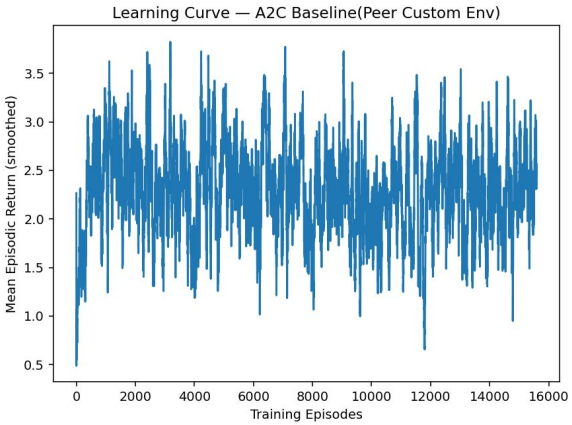
- Results for baseline PPO (the clear winner), A2C and TRPO.
- DQN ran slowly, so despite showing very strong results, we had to stop training.

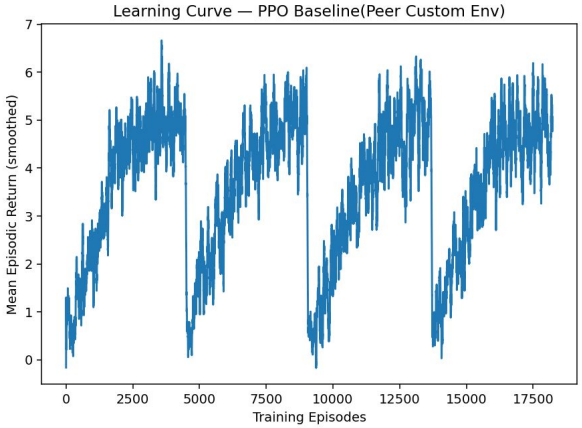


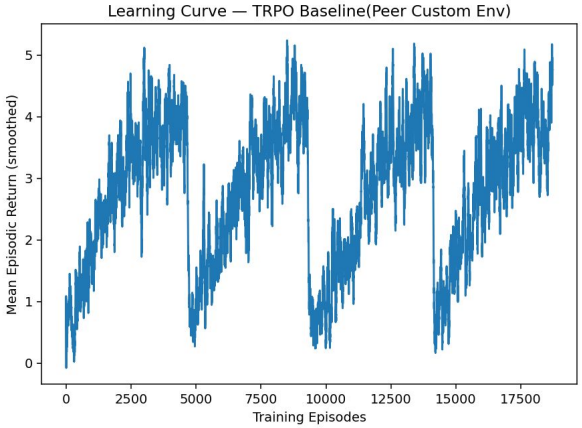

| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| A2C_baseline_1 | 2.4484 | 2.6549 | 204,000 | 5.224 hr |
| DQN_Group5_Complete_Seed42_1 | 5.1908 | 5.2289 | 65,880 | 13.34 hr |
| PPO_baseline_1 | 4.8946 | 5.3227 | 204,800 | 5.507 hr |
| TRPO_baseline_1 | 3.7502 | 3.6778 | 204,800 | 5.125 hr |

# Task 2b : Learning curves for baseline

- A2C, PPO, TRPO side-by-side: A2C most jagged, PPO/TRPO smoother.
- Same env, different algos show behavior differences are algorithmic

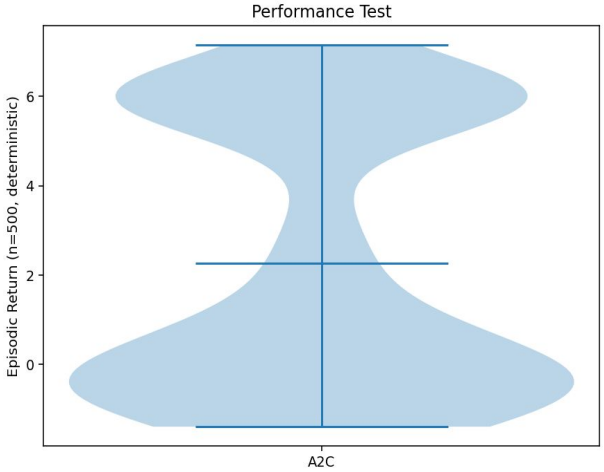


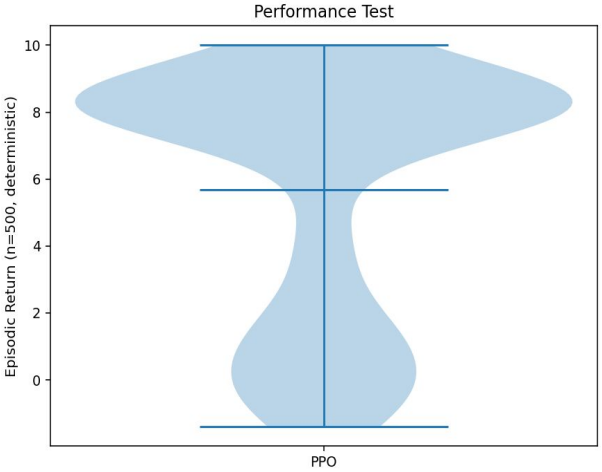



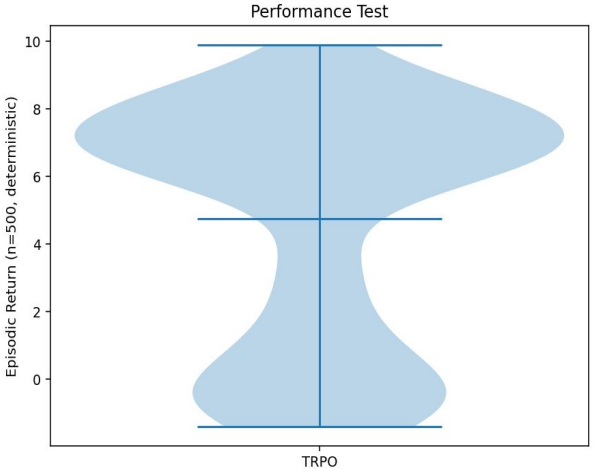



# Task 2b : Performance test for baseline

- PPO/TRPO violins narrower than A2C with fewer bad episodes.
- Same eval protocol; stability advantage shows in distribution



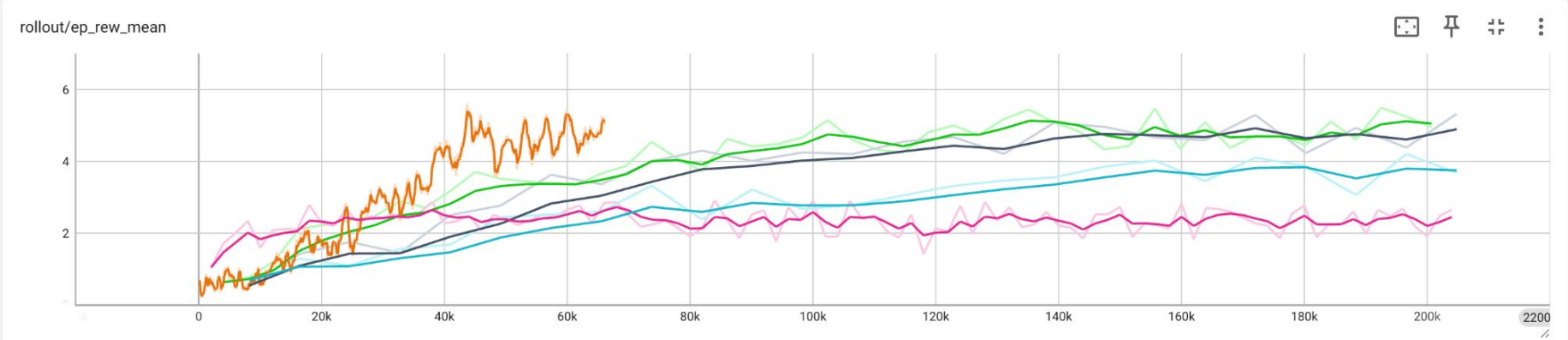



# Task 2b: Optimization of PPO Baseline

- Our baseline models already beat Group 5's own custom env agent. However, we continued to optimize our PPO baseline.
- Here are the things we changed by manual hyperparameter tuning-

| Parameter | Baseline PPO | Improved PPO | Reason for change |
|---|---|---|---|
| n_steps | 4096 | 1024 | Smaller 1024-step rollouts keep data fresh and more representative of current policy |
| batch_size | 64 | 256 | Increased batch size for more stable gradient updates |
| gamma | 0.99 | 0.95 | Smaller gamma emphasizes more immediate rewards - better suited for Grp 5's 50 step episodes |

# Optimized PPO Model for Peer Custom Env

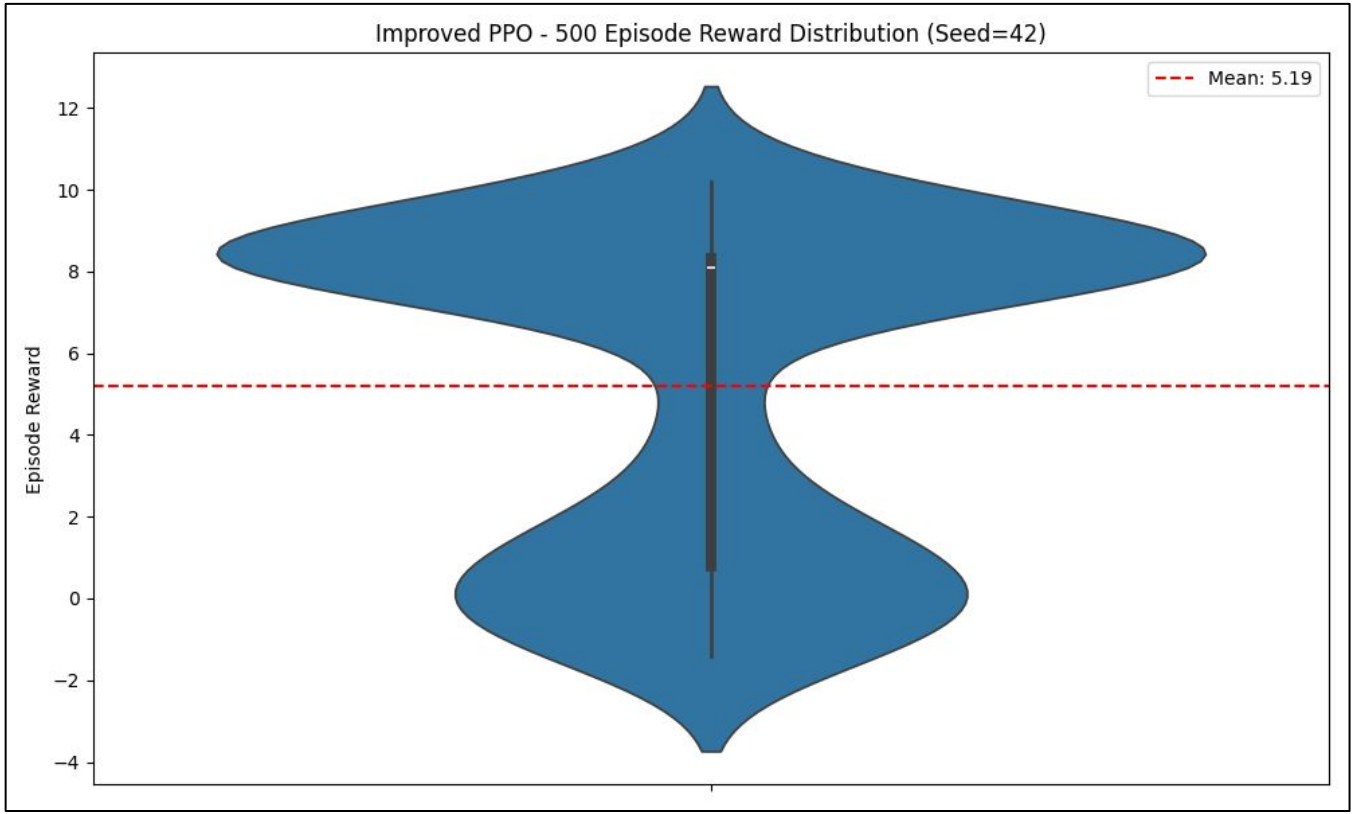



| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| A2C_baseline_1 | 2.4484 | 2.6549 | 204,000 | 5.224 hr |
| DQN_Group5_Complete_Seed42_1 | 5.1017 | 5.1142 | 66,052 | 13.36 hr |
| Improved PPO | 5.0532 | 4.9568 | 200,704 | 4.874 hr |
| PPO_baseline_1 | 4.8946 | 5.3227 | 204,800 | 5.507 hr |
| TRPO_baseline_1 | 3.7502 | 3.6778 | 204,800 | 5.125 hr |

Improved PPO (in green) shows much better learning than the Baseline PPO.

# Performance Test - Optimized PPO Model

**Mean: 5.19**
**Median: 8.11**
**Std Dev.: 4.04**
**Min: -1.40**
**Max: 10.20**
**Q25: 0.78**
**Q75: 8.33**



# THANK YOU